

ENGR 102 Summer Bridge

Day 14 Instructor Key

Lists, Length, Indexes, Mutation, and List Loops

20 points • approximately 20-25 minutes

Name		UIN		Section	
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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace stored state and exact output.

4 points

```
data = [5, 2, 8, 2]
positions = []
total = 0
for index in range(len(data)):
    if data[index] == 2:
        positions.append(index)
    total = total + data[index] * index
print(total, positions)
```

Give an index ledger and exact output.

Answer and explanation

Weighted contributions are $5*0 = 0$, $2*1 = 2$, $8*2 = 16$, and $2*3 = 6$, giving total 24. Values equal to 2 occur at indexes 1 and 3. Exact output: 24 [1, 3].

Question 2. Repair a list-bound error.

3 points

```
data = [4, 7, 9]
for index in range(len(data) + 1):
    print(data[index])
```

State the output before failure, name the failing index, and give the minimal repair.

Answer and explanation

The code prints 4, 7, and 9, then tries `data[3]`. Index 3 is invalid for length 3, so an `IndexError` occurs. Repair with `range(len(data))`.

Question 3. Program construction**13 points**

Write one complete Python program that satisfies every requirement.

- Use `data = [5, 12, 5, 18, 2]`. Do not use `sum`, `max`, or `index`.
- Loop by index. Compute the sum of values stored at even indexes.
- Count all values at or above 10.
- Find the index of the first largest value.
- Print `even_index_sum`, `high_count`, and `largest_index`. Include one tie test.

Answer and explanation

```
data = [5, 12, 5, 18, 2]
even_index_sum = 0
high_count = 0
largest = data[0]
largest_index = 0

for index in range(len(data)):
    value = data[index]
    if index % 2 == 0:
        even_index_sum = even_index_sum + value
    if value >= 10:
        high_count = high_count + 1
    if value > largest:
        largest = value
        largest_index = index

print(even_index_sum)
print(high_count)
print(largest_index)
```

Even indexes 0, 2, and 4 store 5, 5, and 2, so `even_index_sum` is 12. Values 12 and 18 meet the threshold, so `high_count` is 2. The first largest value 18 is at index 3.

Tie test `[9, 9]` should keep `largest_index` 0; using `>` rather than `>=` preserves the first occurrence.

Scoring guidance

3 initialization/traversal, 3 even-index sum, 2 threshold count, 3 first-largest index, 1 output, 1 tie test.